

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOBIDIR

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Sertifikat_pro**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!
PROMOCK–1. 26.07.2026

1. x va y natural sonlar. Agar $x \cdot y = 2400$ va x ning natural bo'luvchilari soni y ning natural bo'luvchilari sonidan 4 marta katta bo'lsa, $x + y$ ning eng kichik qiymatini toping.

A) 110
B) 121
C) 130
D) 140

2. Hisoblang:

$$417 \cdot \left(\frac{2}{10} + \frac{13}{990} \right) : \left(\frac{4}{10} + \frac{21}{990} \right) .$$

A) 111
B) 201
C) 211
D) 210

3. Bir avtomobil soat 12:00 da A shahardan B shaharga yo'lga chiqdi va 17:30 da yetib bordi. Avtobus B shahardan A shaharga 13:00 da yo'lga chiqdi. Ular soat 16:00 da uchrashgan bo'lsa, avtobus A shaharga soat nechada yetib boradi?

A) 21 : 00
B) 22 : 00
C) 22 : 30
D) 00 : 00

4. Qadoqlash kompaniyasida K va L robotlari bir xil turdagi mahsulotlarni qadoqlaydi. L robotining 1 dona mahsulotni qadoqlashi uchun ketadigan vaqt, K robotinikidan 1 daqiqa ko'pdir. K robot 18 daqiqa, L robot esa 12 daqiqa ishlagan va bu ikki robot jami 63 dona mahsulot qadoqlagan. Shunga ko'ra, K va L robotlari bir vaqtda 36 daqiqa ishlaganda jami nechta mahsulot qadoqlangan bo'ladi?

A) 108 ta
B) 120 ta
C) 135 ta
D) 145 ta

5. Hisoblang:

$$(5 \cdot 3)^2 - ((-5) : (-3))^{-2} .$$

A) 224,64
B) 225
C) 224
D) 225,64

6. Hisoblang:

$$\frac{\sqrt{1 + 2026^2 + \frac{2026^2}{2027^2}} + \frac{2026}{2027}}{2027}.$$

- A) 1
- B) 2026
- C) 2027
- D) 4053

7. Hisoblang:

$$\frac{1 + \frac{\sqrt{3}}{2}}{1 + \sqrt{1 + \frac{\sqrt{3}}{2}}} + \frac{1 - \frac{\sqrt{3}}{2}}{1 - \sqrt{1 - \frac{\sqrt{3}}{2}}}.$$

- A) $\frac{1}{2}$
- B) 1
- C) $\sqrt{3}$
- D) 2

8. Arifmetik progressiyada:

$$\begin{cases} S_5 - S_2 - a_5 = 0,1 \\ S_4 + a_7 = 0,1 \end{cases}$$

bo'lsa, a_1 ni toping.

- A) 0,5
- B) -0,5
- C) 0,7
- D) -0,7

9. (b_n) geometrik progressiyasida $b_{11} = \log_5 x$ va $b_{25} = \log_x 625$ bo'lsa, bu ketma-ketlikning 18-hadi nechaga teng?

- A) 2
- B) ± 3
- C) ± 2
- D) 3

10. Soddalashtiring:

$$\frac{x^8 + x^4 + 1}{x^6 - 1} - \frac{x^4 - x^2 - 1}{x^2 - 1}.$$

- A) $\frac{2}{x^2 - 1}$
- B) $\frac{1}{x^2 - 1}$
- C) $\frac{2}{x - 1}$

D) $\frac{2}{x+1}$

11. Bo‘linmani toping va natijani $x = 0, 2$; $y = \frac{1}{2}$; $z = \frac{4}{5}$ bo‘lganda hisoblang:

$$\frac{x+y-z}{x+y+z} : (x^2 - y^2 - z^2 + 2yz) .$$

A) $\frac{3}{4}$

B) $\frac{2}{3}$

C) $\frac{1}{3}$

D) $\frac{4}{3}$

12. Ifodani soddalashtiring:

$$\frac{(\sin x - \cos x)^2}{\sin^2 \left(\frac{\pi}{4} - x \right)} .$$

A) 2

B) $\frac{1 - \sin^2 x}{1 + \sin^2 x}$

C) $\frac{2(1 - \sin^2 x)}{1 - \cos^2 x}$

D) 1

13. $\sqrt{3} \sin 2x - \cos 2x = \sqrt{3}$ tenglama $(0; 5\pi)$ oraliqda nechta ildizga ega?

A) 20 ta

B) 18 ta

C) 10 ta

D) 8 ta

14. Tenglamalar sistemasini yeching:

$$\begin{cases} 2^y \cdot 8^{-x} = 8\sqrt{2} \\ y + 3x = \frac{1}{2} . \end{cases}$$

A) $\left(\frac{1}{2}; 2 \right)$

B) $\left(-\frac{1}{2}; 2 \right)$

C) $\left(\frac{1}{2}; -2 \right)$

D) $\left(-\frac{1}{2}; -2 \right)$

15. Quyidagi tenglamasining haqiqiy ildizlari yig'indisini toping:

$$(x^2 - 2x)^2 + 8x^2 - 16x - 9 = 0.$$

- A) 1
- B) 2
- C) 3
- D) 4

16. Tengsizlikning eng kichik butun musbat va eng katta butun manfiy yechimlari yig'indisini toping:

$$\sqrt{\frac{x+1}{x-1}} - \sqrt{\frac{x-1}{x+1}} \leq \frac{3}{2}.$$

- A) -1
- B) 1
- C) 2
- D) 0

17. Tengsizlikni yeching:

$$\frac{1}{2} \log_{0,1}(6+x) \leq \log_{0,1} x.$$

- A) (0; 3]
- B) [0; 3]
- C) (0; 3)
- D) (0; 3) $\cup$ (3; ∞)

18. Tengsizlik nechta butun yechimga ega:

$$\frac{|x-5|(x^2-x-20)\sqrt{x+2}}{x^2+9x+14} \leq 0.$$

- A) 10 ta
- B) 7 ta
- C) 9 ta
- D) 8 ta

19. Tenglamani yeching:

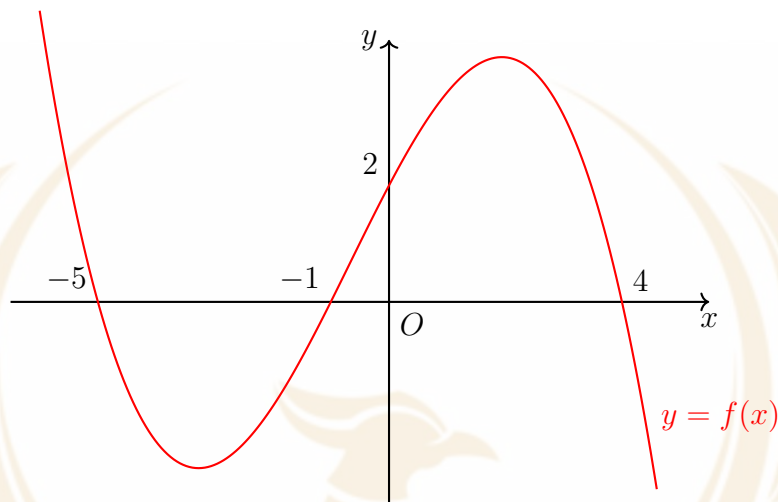
$$\frac{x+23}{25} + \frac{x+30}{19} + \frac{x+27}{54} = 6.$$

- A) 22
- B) 23
- C) 27
- D) 29

20. $x \in (-\infty, 1]$ bo'lsin. $f(x) = x^2 - 2x$ funksiyaning teskarisini toping.

- A) $\sqrt{x+1} + 1$
- B) $1 - \sqrt{x+1}$
- C) $\sqrt{x+1}$
- D) $\sqrt{x-1}$

21. $f(x)$ funksiyasining grafigi quyida berilgan.



Shunga ko'ra, $y = f(2x + 1)$ funksiyasining x o'qini kesib o'tadigan nuqtalari absissalarining yig'indisi nechaga teng?

- A) -3
- B) -2
- C) $-\frac{3}{2}$
- D) $-\frac{5}{2}$

22. $f(x) = \arctg(\cos 2x)$ bo'lsa, $f'\left(\frac{\pi}{8}\right)$ ning qiymatini toping.

- A) $-\frac{4\sqrt{2}}{3}$
- B) $-\frac{\sqrt{2}}{3}$
- C) $-\frac{2\sqrt{2}}{3}$
- D) $\frac{\sqrt{2}}{3}$

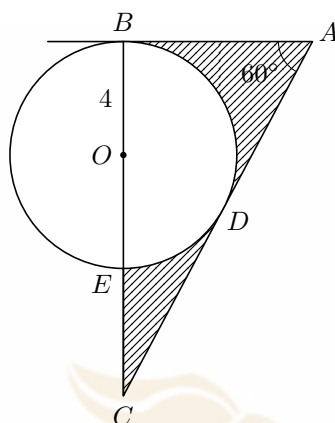
23. Integralni hisoblang:

$$\int_1^2 \frac{2x+1}{x^3+x^2} dx .$$

- A) $\ln \frac{3}{2} - \frac{1}{2}$

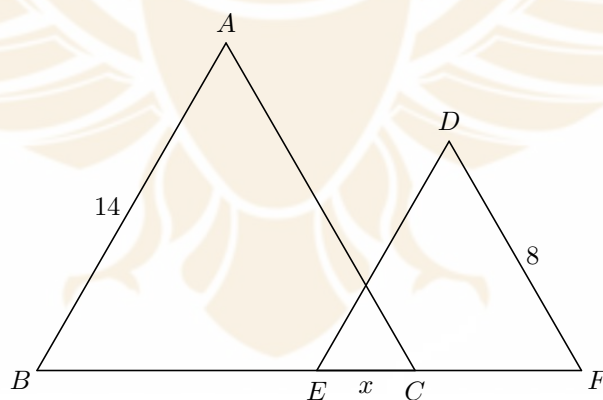
- B) $\ln \frac{3}{2} + \frac{1}{2}$
 C) $\ln \frac{4}{3} - \frac{1}{2}$
 D) $\ln \frac{4}{3} + \frac{1}{2}$

24. AB urinma O markazli aylanaga B nuqtada, AC urinma esa D nuqtada urinadi. $\angle BAC = 60^\circ$ va $OB = 4$ cm berilgan. Yuqoridagi ma'lumotlarga ko'ra, bo'yalgan sohalar yuzining yig'indisini toping?



- A) $24\sqrt{3} - 6\pi$
 B) $24\sqrt{3} - 8\pi$
 C) $24\sqrt{3} - 9\pi$
 D) $12\sqrt{3} - 3\pi$

25. ABC va DEF muntazam uchburchak berilgan. Agar $BF = 16$ cm, $DF = 8$ cm va $AB = 14$ cm bo'lsa, $EC = x$ necha cm ga teng?

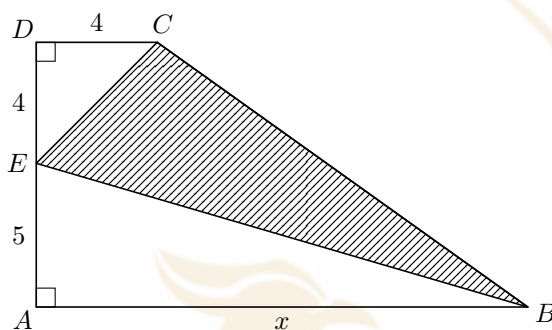


- A) 6
 B) $\frac{9}{2}$
 C) 5
 D) $\frac{11}{2}$

26. ABC teng yonli $AB = BC$ uchburchakka aylana ichki chizilgan va yon tomonini urinish nuqtasidan 2 va 5 ga teng kesmalarga ajratgan bo'lib, agar $AB < AC$ bo'lsa, uchburchak yuzini toping.

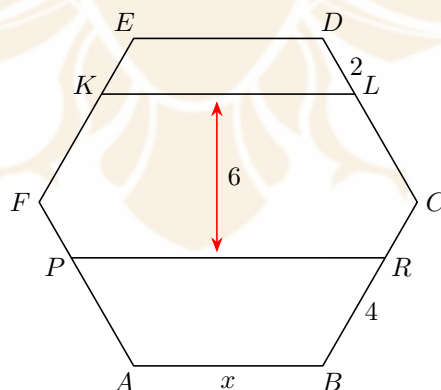
- A) $10\sqrt{3}$
 B) $5\sqrt{6}$
 C) $5\sqrt{3}$
 D) $10\sqrt{6}$

27. $ABCD$ to'g'ri burchakli trapetsiya, $S_{BEC} = 30 \text{ cm}^2$, $DC = DE = 4 \text{ cm}$ va $AE = 5 \text{ cm}$ bo'lsa, AB kesma uzunligini toping? (cm)



- A) 7
 B) 8
 C) 10
 D) 9

28. $ABCDEF$ muntazam oltiburchak, $ED \parallel KL \parallel PR$, $DL = 2 \text{ cm}$ va $RB = 4 \text{ cm}$. KL va PR to'g'ri chiziqlar orasidagi masofa 6 cm ekanligi ma'lum bo'lsa, AB kesma uzunligini toping? (cm)

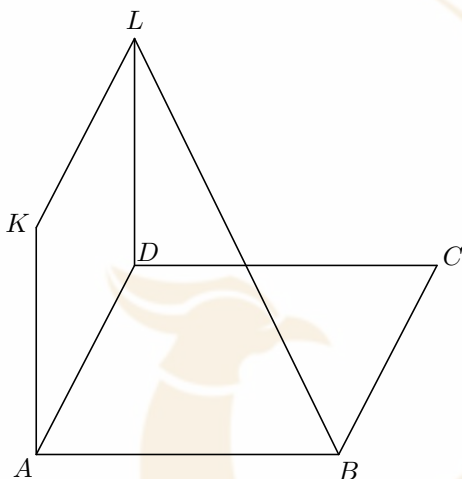


- A) $4 + \sqrt{3}$
 B) $2 + 3\sqrt{3}$
 C) $2 + 2\sqrt{3}$
 D) $3 + 2\sqrt{3}$

29. ABC uchburchak yuzi 120 ga teng. Agar $\overrightarrow{BE} = -\frac{1}{2}\overrightarrow{BA}$ bo'lsa, BEC uchburchak yuzini toping.

- A) 60
- B) 120
- C) 240
- D) 30

30. Tekislikda $ABCD$ kvadrat tekisligi bilan $ADLK$ kvadrat tekisligi o'zaro perpendikulyar, $AB = 8$ sm. Berilganlarga ko'ra, LB uzunligini toping.

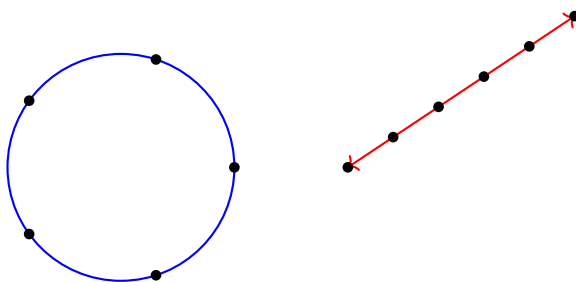


- A) 10
- B) 12
- C) $8\sqrt{2}$
- D) $8\sqrt{3}$

31. $U = \{x \mid -12 \leq x \leq 12, x \in \mathbb{Z}\}$ universal to'plam hamda uning $A = \{x \mid -8 \leq x \leq 4, x \in \mathbb{Z}\}$ va $B = \{x \mid -4 \leq x \leq 8, x \in \mathbb{Z}\}$ qism to'plamlari berilgan. $(A \cap B)'$ to'plamning elementlari sonini toping.

- A) 12
- B) 16
- C) 18
- D) 20

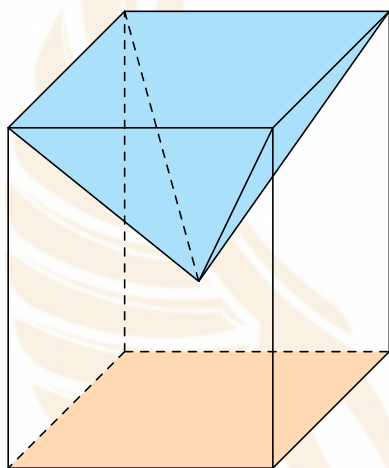
32. Rasmdagi 11 ta nuqtadan 4 tasini tanlab, eng ko'pi bilan nechta har xil to'rtburchak chizish mumkin?



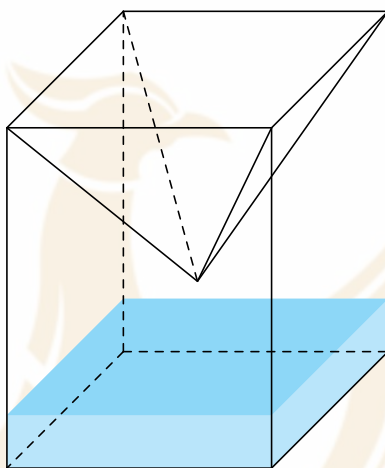
- A) 330
- B) 315
- C) 215
- D) 120

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Quyidagi shakllarda kvadrat prizma ko‘rinishidagi idish ichiga muntazam to‘rtburchakli piramida ko‘rinishidagi idish joylashtirilgan. Piramidaning asosi bilan prizmaning ustki asosi o‘zaro teng va ustma-ust tushadi. 1–shaklda piramida ko‘rinishidagi idish to‘liq suv bilan to‘la bo‘lganda, uning uchi (tepa nuqtasi) tomonida ochilgan teshik yordamida suv to‘liq 2-shakldagi kabi prizmaning ichiga bo‘shatiladi. 1–shaklda piramidaning uchi (tepa nuqtasi)ning prizma asosigacha bo‘lgan masofasi 12 cm, 2–shaklda esa piramidaning uchi (tepa nuqtasi)ning suvning ustki sathigacha bo‘lgan masofasi 8 cmni tashkil qiladi.



1 – chizma



2 – chizma

Javoblar.

- A) $\frac{1}{3}$
- B) $\frac{1}{6}$
- C) 24
- D) $\frac{10}{3}$
- E) 22
- F) $\frac{1}{2}$

33. Bunga ko‘ra, prizmaning balandligi necha cm?

34. Piramida hajmini, prizma hajmiga nisbatini toping.

35. Agar asos diagonali $10\sqrt{2}$ cm bo‘lsa, piramidaga ichki chizilgan shar radiusini toping. (dm)

36. Tenglamani yeching:

$$\sqrt{5 + \sqrt{5 + x}} = x .$$

a) Tenglamaning haqiqiy ildizlari yig'indisini toping.

Javob: _____

b) Tenglamaning haqiqiy ildizlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

37. Quyidagi trigonometrik tenglamani yeching:

$$4 \sin 8x \sin 2x = 4 \sin^2 5x - 3 .$$

a) Tenglamani $0 < x < \pi$ oralig'idagi ildizlari yig'indisini toping.

Javob: _____

b) Tenglamani $0 < x < \pi$ oralig'idagi eng katta va eng kichik ildizlari musbat farqini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

38. $f(x)$ funksiya uchun quyidagi tenglik berilgan:

$$f(x^2 + x) + 2f(x^2 - 3x + 2) = 9x^2 - 15x .$$

a) $f(2027)$ ni toping.

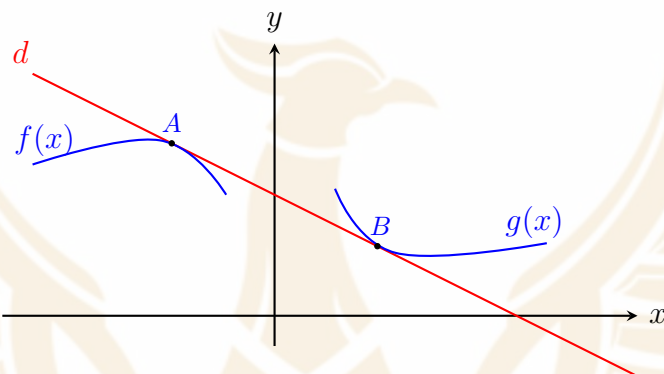
Javob: _____

b) $f(x) = 2$ tenglamani yeching.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

39. $f(x) = -x^3 - x$ va $g(x) = x^3 - 4x + 1$ funksiyalarining grafiklari quyida tasvirlangan. d to'g'ri chizig'ining funksiyalarga urinma bo'lgan A va B urinish nuqtalarining absissalari mos ravishda x_1 va x_2 bo'lsin.



a) $x_1^2 + x_2^2$ ni toping.

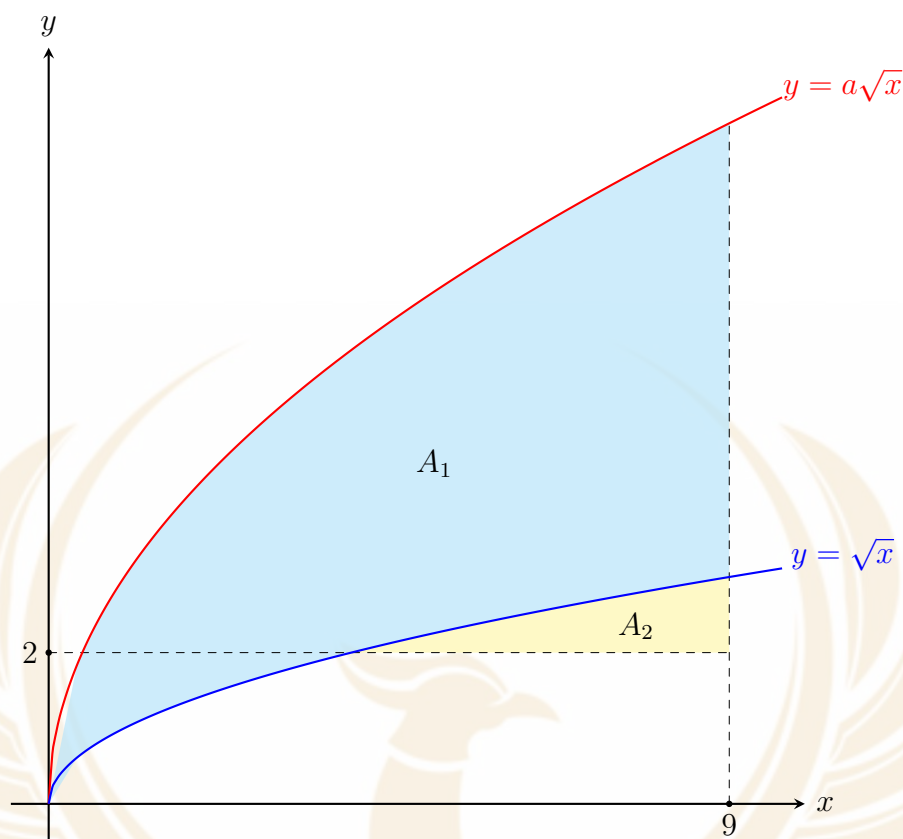
Javob: _____

b) $x_1^3 + x_2^3$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

40. a haqiqiy son bo'lib, dekart koordinatalar sistemasida $y = a\sqrt{x}$ va $y = \sqrt{x}$ funksiyalarining grafiklari quyida tasvirlangan.



Ko'k rangli soha yuzi A_1 , sariq rangli soha yuzi A_2 bo'lib, $A_1 \cdot A_2 = 96$ bo'lsa,

a) a nechaga teng?

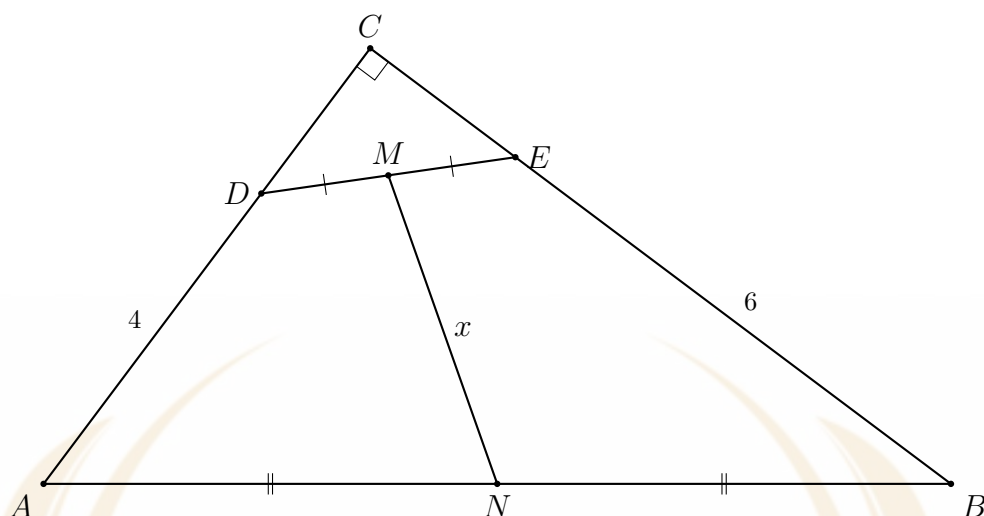
Javob: _____

b) $A_1 - A_2$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. Chizmada ABC to'g'ri burchakli uchburchak tasvirlangan. $AD = 4$ cm, $EB = 6$ cm, $DM = ME$ va $AN = BN$ tengliklar o'rinli.



a) $MN = x$ ning qiymatini toping. (cm)

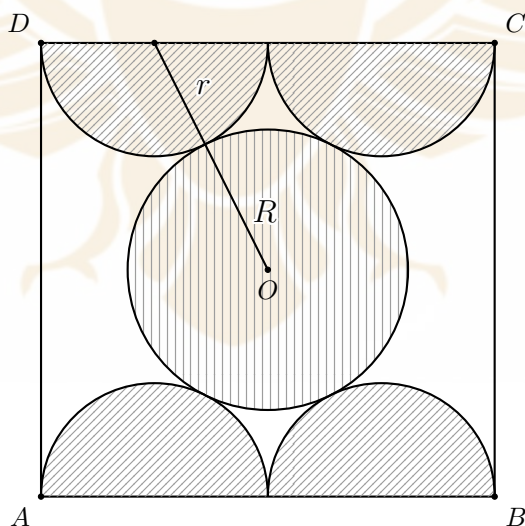
Javob: _____

b) Agar C , M , N nuqtalar bir to'g'ri chiziqda yotsa, $AB - DE$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqarajasiga ko'chirib yozing.

42. Quyidagi rasmda $ABCD$ kvadratga R radiusli doira va to'rtta r radiusli yarim doiralar chizilgan.



a) $\frac{4r}{R}$ ni toping.

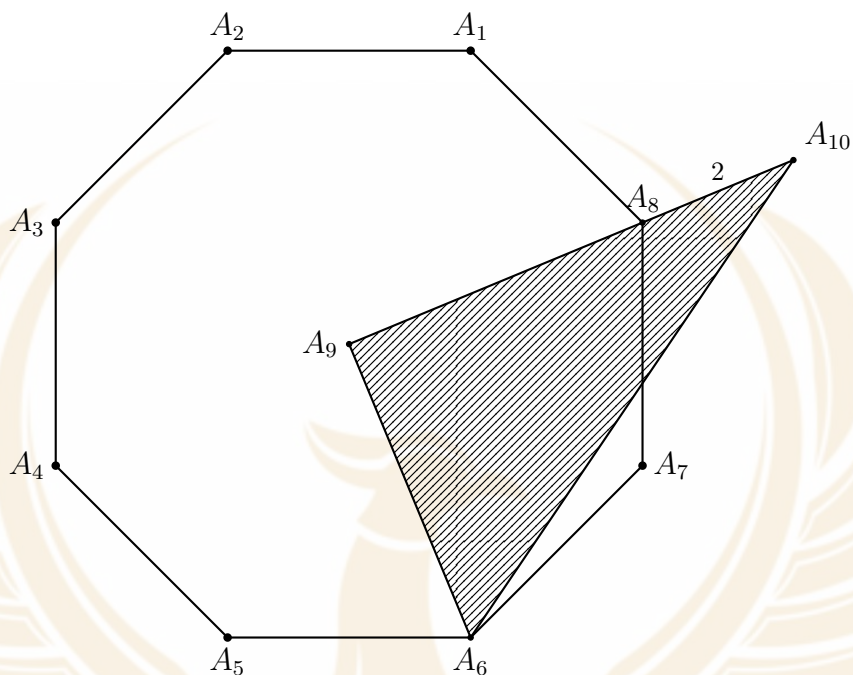
Javob: _____

b) Katta doira yuzini kichik doira yuziga nisbatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ga ko'chirib yozing.

43. $A_1A_2A_3A_4A_5A_6A_7A_8$ muntazam sakkiz burchak va ABC uchburchak rasmdagidek tasvirlangan. $A_6A_{10} = 10$ cm ga, $A_8A_{10} = 2$ cm ga teng.



- a) $A_9A_6A_{10}$ uchburchak yuzini toping. (cm^2)

Javob: _____

- b) Muntazam sakkizburchak yuzini toping. (cm^2)

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ga ko'chirib yozing.

